Research Article

Secrecy Performance of Energy Harvesting Backscatter-Assisted Hybrid Long and Short-Packet Communications

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Abstract. *This paper investigates the secrecy performance of an energy-harvesting backscatter-assisted system supporting hybrid long- and short-packet communications. A source transmits long-packet information to a legitimate destination in the presence of a passive eavesdropper, while an energy-constrained backscatter device harvests energy from the incident radio-frequency signal. Once activated, the device embeds its confidential short-packet information into the incident waveform and reflects it towards the destination. All wireless links are modelled as independent quasi-static Rayleigh fading channels. A closed-form expression is derived for the secrecy outage probability of the long-packet transmission. For the short-packet transmission, the average secure block error rate is analysed by jointly considering the energy-activation requirement, finite-blocklength coding, successive decoding, and information leakage. Since the resulting integrals are analytically intractable, an efficient Gaussian–Chebyshev quadrature approximation is developed. Numerical results closely match Monte Carlo simulations and show that increasing the transmit SNR and blocklength improves secure transmission reliability. The results also demonstrate that performance is governed by the relative qualities of the legitimate and eavesdropping channels, rather than by transmit power alone.*

Keywords

Ambient backscatter communication, energy harvesting, finite-blocklength communication, secrecy outage probability, secure block error rate, short-packet communication.

1. Introduction

The Internet of Things (IoT) has emerged as a key technological paradigm in which sensors, communication devices, and physical objects are interconnected to collect, exchange, and process information with limited human intervention [1]. This connectivity supports a broad range of applications, including smart cities, industrial automation, environmental monitor-

ing, and intelligent healthcare. However, many IoT devices operate with limited battery capacity and are deployed in locations where regular maintenance or battery replacement is difficult. Energy consumption and operational sustainability therefore remain important challenges in the development of large-scale IoT networks [2]. These limitations have encouraged the use of energy harvesting and low-power communication technologies to extend device lifetime while maintaining reliable data transmission [3].

Nevertheless, sustaining the operation of large-scale IoT networks remains challenging because many low-power devices depend on finite-capacity batteries and are deployed in locations where battery replacement is costly or infeasible [4]. Energy harvesting (EH) and wireless power transfer (WPT) provide viable means of addressing this limitation by enabling energy-constrained devices to obtain power from ambient or dedicated radio-frequency (RF) signals. In addition to reducing maintenance requirements, the harvested RF energy can support cooperative communication and low-power transmission architectures, thereby extending device lifetime and enabling more continuous network operation [5]. This capability is particularly attractive for ambient backscatter communication (BackCom), in which a passive device can exploit an incident radio-frequency signal for both energy supply and information transmission. Many researchers focus on the EH in Backcom networks to enhance the reliability of their proposed system. Specifically, the authors investigated the throughput and energy efficiency in a heterogeneous wireless energy harvesting IoT network using an opportunistic Backscatter communication medium access control (OBM) protocol [6]. The work presented in [7] clearly formulates a model for wireless-powered ambient backscatter devices and derives a definitive closed-form expression for outage probability (OP) under Rayleigh fading. Furthermore, the authors establish the power-splitting factor that effectively balances the trade-off between EH and achievable data rate, demonstrating a comprehensive and precise approach to the problem.

Moreover, in many IoT applications, sensing and control devices transmit only a small number of information bits under stringent latency and reliability requirements. Short-packet communication is therefore well suited to applications such as industrial automation, environmental monitoring, and event-driven sensing. However, the classical Shannon capacity is derived under the assumptions of infinitely long codewords and a vanishing decoding error probability, and may consequently provide an inaccurate performance characterisation when the blocklength is finite [8, 9]. Under the finite-blocklength regime, the maximum achievable coding rate depends not only on the instantaneous channel capacity, but also on the blocklength, target decoding error probability, and channel dispersion. A shorter blocklength reduces the transmission delay, but generally increases the finite-blocklength penalty and the probability of decoding failure, giving rise to a fundamental trade-off among latency, reliability, and energy efficiency in short-packet IoT communications [10–12]. On the other hand, the communication networks integrated hybrid long and short packets were introduced in [13–15]. In [13], the expressions for the primary transmitter (PT)'s outage probability (OP) and the backscatter transmitter (BT)'s average block error rate (BLER) were definitively derived by introducing an improved successive interference cancellation (SIC) decoding scheme. Furthermore, [14] rigorously analyzed the performance of the BackCom network employing hybrid automatic repeat request (HARQ) with hybrid long and short packets, where the base station transmits long packets to a cellular user while an IoT node executes short-packet BackCom with retransmissions. Finally, Jiayao Shi and colleagues looked into how an ambient backscatter communication (AmBackCom) network performs when it handles both long and short packets. In their setup, long packets are used for cellular transmission, while several IoT nodes take turns sending short packets through Backcom [15]. They managed to come up with exact formulas for the BLER of each IoT node's backscatter transmission, along with upper and lower bounds for the OP in the cellular transmission.

In addition, Physical-layer security (PLS) has emerged as a promising complement to conventional cryptographic methods by exploiting the inherent randomness of wireless channels to protect confidential information from unauthorised receivers [16, 17]. In quasi-static fading environments, however, the instantaneous secrecy capacity may fall below the target secrecy rate when the legitimate channel is not sufficiently stronger than the eavesdropping channel. Secrecy outage probability (SOP), defined as the probability of this event, is therefore widely employed to quantify the secrecy reliability of wireless communication systems [18, 19]. This metric is particularly relevant to energy-constrained and cooperative networks because energy harvesting, channel fading, relaying strategies, and resource-allocation parameters jointly affect the quality advantage of the legitimate link. Accordingly, analysing SOP provides both a tractable measure of confidentiality and practical guidance for designing secure wireless-powered communication systems.

Other aspects of PLS, in secure short-packet communications, conventional SOP alone may not provide a sufficiently comprehensive performance characterisation because finite-length codewords introduce a non-negligible decoding error even when the instantaneous secrecy capacity exceeds the target cod-

ing rate. Secure block error rate (SBLER) is therefore employed to jointly account for transmission reliability and information confidentiality under finite-blocklength constraints. In the considered energy-constrained backscatter system, a secure packet delivery failure may result from insufficient harvested energy to activate the BD, unsuccessful decoding at the legitimate destination, or failure to satisfy the prescribed information-leakage constraint against the eavesdropper. Accordingly, SBLER offers a unified measure that incorporates energy availability, coding rate, blocklength, channel dispersion, decoding reliability, and secrecy leakage [20, 21]. This metric is especially appropriate for low-latency IoT applications, in which confidential information must be transmitted using short packets while maintaining stringent reliability and security requirements [22–24].

Unlike the aforementioned studies, this work jointly investigates the secrecy performance of long-packet transmission and secure short-packet transmission in an energy-harvesting ambient backscatter communication system. The main contributions of this paper are summarised as follows:

- We propose a hybrid communication framework in which an energy-constrained backscatter device operates according to the instantaneous harvested energy. The system switches between long-packet-only transmission and hybrid long- and short-packet transmission depending on whether the backscatter device can be activated.
- We derive a closed-form expression for the secrecy outage probability of the long-packet transmission over quasi-static Rayleigh fading channels, thereby characterising the effects of the transmit SNR, target secrecy rate, and legitimate and eavesdropping channel conditions.
- We formulate the secrecy block error rate of the short-packet transmission under finite blocklength and derive its average value by jointly considering energy activation failure, decoding errors, and information leakage.
- To address the analytically intractable integrals arising in the average secrecy block error rate, we develop a Gaussian–Chebyshev quadrature approximation that enables accurate and efficient numerical evaluation.
- Monte Carlo simulations verify the analytical results and illustrate the effects of the principal system parameters on secrecy and transmission reliability.

The remainder of this paper is organised as follows. Section 2 introduces the system and channel models. Section 3 presents the secrecy outage probability and average secrecy block error rate analyses. Section 4 provides the numerical results and corresponding discussion. Finally, Section 5 concludes the paper.

2. System Model

As illustrated in Fig. 1, we consider a hybrid long- and short-packet wireless communication system comprising a source S, an energy-constrained backscatter device BD, a legitimate destination D, and a passive eavesdropper E. Each node is equipped with a single antenna. The source transmits long-packet information to D over the direct link S $\rightarrow$ D. Owing to the broadcast nature of wireless communication, the long-packet signal may also be intercepted by E through the wiretap link S $\rightarrow$ E.

In addition to carrying the long-packet information, the radio-frequency signal transmitted by S serves as an ambient carrier and an energy source for BD. Specifically, BD receives the incident signal through the S $\rightarrow$ BD link and harvests energy from this signal to power its backscatter circuitry. Since BD has no active radio-frequency transmitter, its operation depends on whether the harvested energy is sufficient to activate the device.

Let h_{ij} denote the complex channel coefficient of the $i \rightarrow j$ link, where

$$(i,j) \in \mathcal{L} \triangleq \{(\mathrm{S},\mathrm{B}),(\mathrm{S},\mathrm{D}),(\mathrm{S},\mathrm{E}),(\mathrm{B},\mathrm{D}),(\mathrm{B},\mathrm{E})\}. \quad (1)$$

Here, B denotes the backscatter device when it is used as a subscript. Accordingly, h_{SB}, h_{SD}, h_{SE}, h_{BD}, and h_{BE} represent the channels associated with the energy-harvesting, direct long-packet, long-packet wiretap, legitimate backscatter, and backscatter wiretap links, respectively.

Based on Fig. 1, the average received power at BD can be expressed as [25]

$$P_h = (1-\beta)\, P_{\mathrm{S}} |h_{\mathrm{SB}}|^2, \quad (2)$$

where $0 < \beta \leq 1$ is the power reflection coefficient, P_{S} is the transmit power of source S.

The operating state of BD is determined by the instantaneous channel power gain of the S $\rightarrow$ BD link. Let $\tau_{\mathrm{EH}} > 0$ denote the prescribed energy-activation threshold. The activity indicator of BD is defined as

$$\mathcal{I}_{\mathrm{BD}} = \begin{cases} 1, & |h_{\mathrm{SB}}|^2 \geq \tau_{\mathrm{EH}}, \\ 0, & |h_{\mathrm{SB}}|^2 < \tau_{\mathrm{EH}}. \end{cases} \quad (3)$$

where $\tau_{\mathrm{EH}} = \frac{P_{\mathrm{th}}}{P_{\mathrm{S}}(1-\beta)} = \frac{\phi}{\Psi(1-\beta)}$ is the saturation threshold of the average rechargeable power of the hardware circuit, while $\Psi = \frac{P_{\mathrm{S}}}{N_0}$ and $\phi = \frac{P_{\mathrm{th}}}{N_0}$.

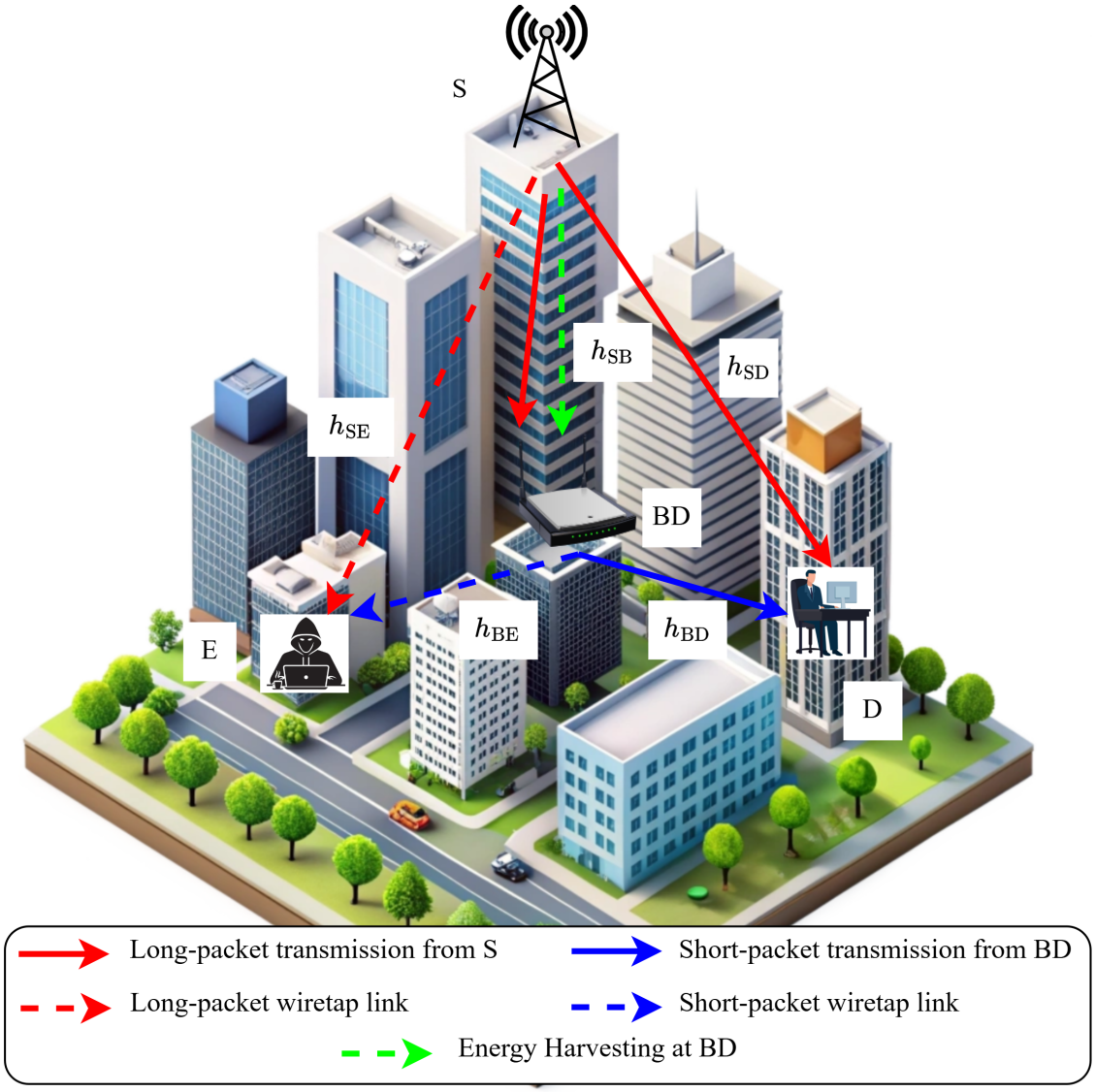


Fig. 1: The considered hybrid long- and short-packet communication network model, comprising a single-antenna source S, a single-antenna backscatter device BD, a single-antenna legitimate destination D, and a single-antenna eavesdropper E.

When $|h_{\mathrm{SB}}|^2 \geq \tau_{\mathrm{EH}}$, the incident radio-frequency signal provides sufficient energy to activate the backscatter circuitry. In this case, BD modulates its short-packet information onto the incident long-packet waveform by varying its antenna load impedance, and subsequently reflects the modulated signal towards D. The resulting short-packet signal is received by D through the BD $\rightarrow$ D link. Meanwhile, E may intercept the same backscattered signal through the BD $\rightarrow$ E wiretap link. Therefore, in the active state of BD, both D and E observe a superposition of the long-packet signal transmitted directly by S and the short-packet signal reflected by BD. In contrast, when $|h_{\mathrm{SB}}|^2 < \tau_{\mathrm{EH}}$, the energy harvested by BD is insufficient to support backscatter operation. Consequently, BD remains inactive and does not transmit its short-packet information. Under this condition, D receives only the long-packet transmission from S, whereas E may overhear the same long-packet transmission. Hence, the proposed system adaptively switches between a hybrid long–short-packet mode and a conventional long-packet-only mode according to the instantaneous amount of energy available at BD.

2.1. Channel Model

All wireless links are assumed to experience quasi-static Rayleigh fading. Thus, the complex channel coefficient of the $i \rightarrow j$ link is modelled as

$$h_{ij} \sim \mathcal{CN}\left(0, \Omega_{ij}\right), \qquad (i,j) \in \mathcal{L}, \tag{4}$$

where

$$\Omega_{ij} = \mathbb{E}\left\{|h_{ij}|^2\right\} \tag{5}$$

denotes the average channel power gain of the corresponding link. The channel coefficients are assumed to remain constant over one transmission block and to vary independently between different blocks. Define the instantaneous channel power gain as

$$\gamma_{ij} \triangleq |h_{ij}|^2 . \tag{6}$$

Since h_{ij} follows a circularly symmetric complex Gaussian distribution, its envelope $|h_{ij}|$ follows a Rayleigh distribution, whereas γ_{ij} follows an exponential distribution with mean Ω_{ij}. Therefore, the probability density function (PDF) of γ_{ij} is given by

$$f_{\gamma_{ij}}(x) = \frac{1}{\Omega_{ij}} \exp\left(-\frac{x}{\Omega_{ij}}\right), \qquad x \geq 0, \tag{7}$$

and $f_{\gamma_{ij}}(x) = 0$ for $x < 0$. The corresponding cumulative distribution function (CDF) is

$$F_{\gamma_{ij}}(x) = 1 - \exp\left(-\frac{x}{\Omega_{ij}}\right), \qquad x \geq 0. \tag{8}$$

Equivalently, by defining the exponential rate parameter as

$$\lambda_{ij} \triangleq \frac{1}{\Omega_{ij}}, \tag{9}$$

Therefore, the PDF and CDF can be rewritten as

$$f_{\gamma_{ij}}(x) = \lambda_{ij} \exp\left(-\lambda_{ij} x\right), \qquad x \geq 0, \tag{10}$$

$$F_{\gamma_{ij}}(x) = 1 - \exp\left(-\lambda_{ij} x\right), \qquad x \geq 0. \tag{11}$$

2.2. Long-Packet Transmission Model

When the channel power gain of the S $\rightarrow$ BD link is below the prescribed energy-activation threshold, i.e., $\gamma_{\mathrm{SB}} < \tau_{\mathrm{EH}}$, the energy harvested by the BD is insufficient to activate its backscatter circuitry. Consequently, the BD remains inactive and does not backscatter its short-packet information. In this case, the system operates in the long-packet-only mode, in which the destination D receives only the long-packet signal directly transmitted by the source S, while the eavesdropper E may overhear the same signal through the S $\rightarrow$ E wiretap link. The received signal at D and E can be given by, respectively.

$$\begin{aligned} y_{\mathrm{D}}^{L} &= \sqrt{P_{\mathrm{S}}} h_{\mathrm{SD}} x_{\mathrm{S}}^{L} + n_{\mathrm{D}}, \\ y_{\mathrm{E}}^{L} &= \sqrt{P_{\mathrm{S}}} h_{\mathrm{SE}} x_{\mathrm{S}}^{L} + n_{\mathrm{E}}, \end{aligned} \tag{12}$$

where x_{S}^{L} denotes long-packet symbol transmitted by S, $\mathbb{E}\left\{\left|x_{\mathrm{S}}^{L}\right|^2\right\} = 1$, whereas $\mathbb{E}$ denotes expectation operator, n_{D} and n_{E} are the additive white Gaussian noise (AWGN) with zero mean and same variance N_0.

Based on (12), the signal-to-noise ratios (SNRs) at D and E are computed as, respectively.

$$\begin{aligned} \gamma_{\mathrm{D}}^{L} &= \frac{P_{\mathrm{S}}|h_{\mathrm{SD}}|^2}{N_0} = \Psi \gamma_{\mathrm{SD}}, \\ \gamma_{\mathrm{E}}^{L} &= \Psi \gamma_{\mathrm{SE}}. \end{aligned} \tag{13}$$

Accordingly, the instantaneous secrecy capacity of the long-packet transmission is given by [26]

$$\begin{aligned} C_{\mathrm{S}}^{L} &= \left[\log_2\left(1+\gamma_{\mathrm{D}}^{L}\right) - \log_2\left(1+\gamma_{\mathrm{E}}^{L}\right)\right]^{+} \\ &= \left[\log_2\left(\frac{1+\gamma_{\mathrm{D}}^{L}}{1+\gamma_{\mathrm{E}}^{L}}\right)\right]^{+}, \end{aligned} \tag{14}$$

where $[x]^{+} \triangleq \max\{x, 0\}$.

For a prescribed target secrecy rate R_{S}^{L}, a secrecy outage event occurs when the instantaneous secrecy capacity is below R_{S}^{L}. Therefore, conditioned on the inactive state of the BD, the secrecy outage probability (SOP) of the long-packet transmission is defined as [27]

$$\begin{aligned} P_{\mathrm{out}}^{L} &\triangleq \Pr\left\{C_{\mathrm{S}}^{L} < R_{\mathrm{S}}^{L}\right\} \\ &= \Pr\left\{\frac{1+\gamma_{\mathrm{D}}^{L}}{1+\gamma_{\mathrm{E}}^{L}} < 2^{R_{\mathrm{S}}^{L}}\right\}. \end{aligned} \tag{15}$$

2.3. Short-Packet Transmission Model

When the channel power gain of the S $\rightarrow$ BD link exceeds the prescribed energy-activation threshold, i.e., $\gamma_{\mathrm{SB}} \geq \tau_{\mathrm{EH}}$, the energy harvested by BD is sufficient to activate its backscatter circuitry. In this case, BD modulates its short-packet information onto the incident long-packet signal transmitted by S by varying its reflection coefficient. The resulting signal is then backscattered towards the legitimate destination D. Owing to the broadcast nature of backscatter communication, the eavesdropper E may also intercept the short-packet signal through the BD $\rightarrow$ E wiretap link.

Let x_{B}^{S} denote the short-packet symbol transmitted by BD, where $\mathbb{E}\left\{\left|x_{\mathrm{B}}^{S}\right|^2\right\} = 1$. The composite signal received at D and E can then be expressed as, respectively.

$$\begin{aligned} y_{\mathrm{D}}^{S} = & \underbrace{\sqrt{P_{\mathrm{S}}} h_{\mathrm{SD}} x_{\mathrm{S}}^{L}}_{\mathrm{direct\,long-packet\,signal}} + \underbrace{\sqrt{\beta P_{\mathrm{S}}} h_{\mathrm{SB}} h_{\mathrm{BD}} x_{\mathrm{S}}^{L} x_{\mathrm{B}}^{S}}_{\mathrm{backscattered\,short-packet\,signal}} \\ & + n_{\mathrm{D}}, \\ y_{\mathrm{E}}^{S} = & \underbrace{\sqrt{P_{\mathrm{S}}} h_{\mathrm{SE}} x_{\mathrm{S}}^{L}}_{\mathrm{direct\,long-packet\,signal}} + \underbrace{\sqrt{\beta P_{\mathrm{S}}} h_{\mathrm{SB}} h_{\mathrm{BE}} x_{\mathrm{S}}^{L} x_{\mathrm{B}}^{S}}_{\mathrm{backscattered\,short-packet\,signal}} \\ & + n_{\mathrm{E}}. \end{aligned} \tag{16}$$

The decoding procedure is performed in two successive stages. In the first stage, nodes D and E decode the long-packet signal transmitted by S while treating

the backscattered short-packet signal as interference. Accordingly, the instantaneous signal-to-interference-plus- noise ratio (SINR) for decoding the long packet at D and E is calculated by, respectively.

$$\begin{aligned}\gamma_{\mathrm{D}}^{L} &= \frac{\Psi|h_{\mathrm{SD}}|^2}{\beta\Psi|h_{\mathrm{SB}}|^2|h_{\mathrm{BD}}|^2+1} \approx \frac{\gamma_{\mathrm{SD}}}{\beta\gamma_{\mathrm{SB}}\gamma_{\mathrm{BD}}},\\ \gamma_{\mathrm{E}}^{L} &= \frac{\Psi|h_{\mathrm{SE}}|^2}{\beta\Psi|h_{\mathrm{SB}}|^2|h_{\mathrm{BE}}|^2+1} \approx \frac{\gamma_{\mathrm{SE}}}{\beta\gamma_{\mathrm{SB}}\gamma_{\mathrm{BE}}}.\end{aligned} \tag{17}$$

After successfully decoding x_{S}^{L}, node j reconstructs and removes the direct long-packet component from the composite received signal by applying successive interference cancellation (SIC). Assuming perfect SIC, the corresponding instantaneous signal-to-noise ratio (SNR) for decoding the short packet at nodes D and E is

$$\begin{aligned}\gamma_{\mathrm{D}}^{S} &= \beta\Psi\gamma_{\mathrm{SB}}\gamma_{\mathrm{BD}},\\ \gamma_{\mathrm{E}}^{S} &= \beta\Psi\gamma_{\mathrm{SB}}\gamma_{\mathrm{BE}}.\end{aligned} \tag{18}$$

Finally, when BD is activated, successful recovery of its short-packet information requires each receiving node to decode the long packet first, cancel the decoded long-packet component via SIC, and then decode the short packet. Therefore, the effective SINRs for jointly decoding the long and short packets at D and E are respectively defined as

$$\begin{aligned}\gamma_{\mathrm{D}} &= \min\left(\gamma_{\mathrm{D}}^{L}, \gamma_{\mathrm{D}}^{S}\right) = \min\left(\frac{\gamma_{\mathrm{SD}}}{\beta\gamma_{\mathrm{SB}}\gamma_{\mathrm{BD}}}, \beta\Psi\gamma_{\mathrm{SB}}\gamma_{\mathrm{BD}}\right),\\ \gamma_{\mathrm{E}} &= \min\left(\gamma_{\mathrm{E}}^{L}, \gamma_{\mathrm{E}}^{S}\right) = \min\left(\frac{\gamma_{\mathrm{SE}}}{\beta\gamma_{\mathrm{SB}}\gamma_{\mathrm{BE}}}, \beta\Psi\gamma_{\mathrm{SB}}\gamma_{\mathrm{BE}}\right).\end{aligned} \tag{19}$$

The minimum operation reflects the successive decoding procedure, since the short packet can be recovered only after the long packet has been successfully decoded and cancelled. Hence, the weaker of the two decoding stages determines the overall reliability of joint long- and short-packet reception.

In this operating mode, we investigate the secure block error rate (SBLER) under the finite-blocklength regime. Specifically, we consider the transmission of a confidential short packet containing B information bits to the legitimate destination D over a blocklength of $L \geq 100$ channel uses. The corresponding coding rate is

$$R_{\mathrm{S}} = \frac{B}{L} \quad \text{bits/channel use.} \tag{20}$$

Under the finite-blocklength regime, the maximum achievable secrecy rate at D, subject to a decoding error probability Φ_{D} and an information-leakage constraint φ, can be approximated as [21, 28]

$$r_{\mathrm{D}} = \begin{cases} C_{\mathrm{S}} - Q^{-1}\left(\Phi_{\mathrm{D}}\right)\sqrt{\frac{V_{\mathrm{D}}}{L}} - Q^{-1}\left(\varphi\right)\sqrt{\frac{V_{\mathrm{E}}}{L}}, & \gamma_{\mathrm{D}} > \gamma_{\mathrm{E}} \\ 0 & , \gamma_{\mathrm{D}} \leq \gamma_{\mathrm{E}} \end{cases} \tag{21}$$

where $C_{\mathrm{S}} = \log_2\left(1+\gamma_{\mathrm{D}}\right) - \log_2\left(1+\gamma_{\mathrm{E}}\right)$ denotes the instantaneous secrecy capacity. Furthermore, V_{D} and V_{E} represent the channel dispersion of the main and wiretap links, respectively, and are given by $V_{\mathrm{D}} = \left(1 - \frac{1}{(1+\gamma_{\mathrm{D}})^2}\right)\left(\log_2\left(e\right)\right)^2$ and $V_{\mathrm{E}} = \left(1 - \frac{1}{(1+\gamma_{\mathrm{E}})^2}\right)\left(\log_2\left(e\right)\right)^2$, and $Q^{-1}(\cdot)$ denotes the inverse of the Gaussian Q-function.

The first dispersion penalty in (21) accounts for the reliability requirement at D, whereas the second accounts for the information-leakage constraint imposed on E. Therefore, secure short-packet transmission is feasible when the achievable secrecy rate satisfies

$$r_{\mathrm{D}} \geq R_{\mathrm{s}} = \frac{B}{L}. \tag{22}$$

Otherwise, a secure block error event occurs. Accordingly, for a given instantaneous channel realisation, the SBLER at the legitimate destination D can be approximated by inverting the finite-blocklength secrecy-rate expression as

$$\Phi_{\mathrm{D}} \approx Q\left(\frac{C_{\mathrm{S}} - Q^{-1}\left(\varphi\right)\sqrt{\frac{V_{\mathrm{E}}}{L}} - R_{\mathrm{s}}}{\sqrt{\frac{V_{\mathrm{D}}}{L}}}\right). \tag{23}$$

Equation (23) applies when $\gamma_{\mathrm{D}} > \gamma_{\mathrm{E}}$. When $\gamma_{\mathrm{D}} \leq \gamma_{\mathrm{E}}$, a positive secrecy rate cannot be supported, and the corresponding SBLER is set to unity. Therefore, the instantaneous SBLER can be written more precisely as

$$\Phi_{\mathrm{D}} \approx \begin{cases} Q\left(\frac{C_{\mathrm{S}} - Q^{-1}\left(\varphi\right)\sqrt{\frac{V_{\mathrm{E}}}{L}} - R_{\mathrm{s}}}{\sqrt{\frac{V_{\mathrm{D}}}{L}}}\right), & \gamma_{\mathrm{D}} > \gamma_{\mathrm{E}}, \\ 1, & \gamma_{\mathrm{D}} \leq \gamma_{\mathrm{E}}. \end{cases} \tag{24}$$

It can be observed from (24) that the instantaneous SBLER depends on the coding rate, blocklength, instantaneous secrecy capacity, information-leakage constraint, and channel dispersions of both the legitimate and wiretap links. Averaging the instantaneous SBLER over all channel realisations yields the average SBLER at D, which is expressed as [20]

$$\varepsilon_{\mathrm{D}} = \mathbb{E}_{\boldsymbol{\gamma}}\left\{\Phi_{\mathrm{D}}\right\}, \tag{25}$$

where $\boldsymbol{\gamma}$ denotes the collection of instantaneous channel power gains involved in γ_{D} and γ_{E}.

To evaluate the overall system performance, the average SBLER, denoted by ε_{D} is obtained by taking the expectation of Φ_{D} over the independent channel fading distributions of γ_{D} and γ_{E}. Thus, ε_{D} is rewitten as

$$\varepsilon_{\mathrm{D}} = \int_{0}^{\infty}\int_{y}^{\infty} \Phi_{\mathrm{D}}(x,y) f_{\gamma_{\mathrm{D}}}(x) f_{\gamma_{\mathrm{E}}}(y)\, dx\, dy. \tag{26}$$

where $f_{\gamma_{\rm D}}(x)$ and $f_{\gamma_{\rm E}}(y)$ denote the PDFs of $\gamma_{\rm D}$ and $\gamma_{\rm E}$, respectively. Note that the lower limit of the inner integral is set to y to strictly enforce the positive secrecy capacity condition ($x > y$).

Due to the analytical intractability of integrating the Q-function over fading distributions, we adopt a first-order linear approximation [29]. Specifically, for given channel realizations $x = \gamma_{\rm D}$ and $y = \gamma_{\rm E}$, the instantaneous error probability $\Phi_{\rm D}(x, y)$ can be approximated as:

$$\Phi_{\rm D}(x,y) = Q\left(\frac{C_{\rm S} - Q^{-1}(\varphi)\sqrt{\frac{V_{\rm E}(y)}{L}} - r_{\rm D}}{\sqrt{\frac{V_{\rm D}(x)}{L}}}\right) \approx \Omega(x,y), \tag{27}$$

where $\Omega(x, y)$ is a piecewise-linear function determined as:

$$\Omega(x,y) = \begin{cases} 1, & x < \vartheta \\ \frac{1}{2} - \nu\sqrt{L}(x - \bar{\phi}), & \vartheta \le x \le \xi \\ 0, & x > \xi \end{cases} \tag{28}$$

And the following terms are defined: $\bar{\phi} = 2^{Q^{-1}(\varphi)\sqrt{\frac{V_{\rm E}(y)}{L}} + r}(1+y) - 1$, $r_{\rm D} = \frac{B}{L}$, $\nu = \frac{1}{\sqrt{2\pi\bar{\phi}(\bar{\phi}+2)}}$, $\vartheta = \bar{\phi} - \frac{1}{2\nu\sqrt{L}}$, and $\xi = \bar{\phi} + \frac{1}{2\nu\sqrt{L}}$.

At high-SNR regimes, the channel dispersion of the eavesdropper link approaches unity ($V_E(y) \approx 1$). Consequently, the center point can be approximated as $\bar{\phi} \approx \omega(1+y) - 1$, wherein $\omega = 2^{\frac{\log_2 e}{\sqrt{L}} Q^{-1}(\varphi) + r_{\rm D}}$. By adopting a step-function approximation, an instantaneous secrecy block error occurs whenever the destination's SINR ($\gamma_{\rm D}$) falls strictly below this threshold.

In this paper, the overall average SBLER ($\epsilon_{\rm D}$) accounts for two independent failure events: (i) the energy outage when the BD fails to harvest sufficient energy, and (ii) the information outage due to decoding errors or secrecy leakage during the short-packet transmission. Hence, by using the law of total probability, $\epsilon_{\rm D}$ can be figured out:

$$\begin{aligned} &\varepsilon_{\rm D} = \Pr(\gamma_{\rm SB} < \tau_{\rm EH}) \\ &+ \Pr(\gamma_{\rm D} < \omega(1+\gamma_{\rm E}) - 1, \gamma_{\rm SB} \ge \tau_{\rm EH}) \\ &= \int_0^{\tau_{\rm EH}} f_{\gamma_{\rm SB}}(x)dx + \\ &\int_{\tau_{\rm EH}}^{+\infty} \Pr(\gamma_{\rm D} < \omega(1+\gamma_{\rm E}) - 1 \mid \gamma_{\rm SB} = x) f_{\gamma_{\rm SB}}(x)dx \end{aligned} \tag{29}$$

3. Performance Analysis

In this section, we derive an analytical framework to evaluate the SOP and the average SBLER of the hybrid long-short packet transmission network. The analysis is consistent with the block Rayleigh fading assumption, wherein the long-packet and short-packet transmission model was presented in Section 2. .

3.1. Secrecy Outage Probability (SOP) Analysis

Based on (13) and (15), the SOP can be computed by

$$\begin{aligned} P_{\rm out}^{L} &= \Pr\left\{\frac{1+\Psi\gamma_{\rm SD}}{1+\Psi\gamma_{\rm SE}} < \gamma_{\rm th}\right\} \\ &= \Pr\left\{\Psi\gamma_{\rm SD} < \tilde{\gamma}_{\rm th} + \gamma_{\rm th}\Psi\gamma_{\rm SE}\right\}, \end{aligned} \tag{30}$$

where $\gamma_{\rm th} = 2^{R_{\rm S}^L}$ and $\tilde{\gamma}_{\rm th} = \gamma_{\rm th} - 1$.

From (30), the SOP is formulated as

$$\begin{aligned} P_{\rm out}^{L} &= \int_0^{+\infty} \Pr\left(\gamma_{\rm SD} < \tfrac{\tilde{\gamma}_{\rm th}}{\Psi} + \gamma_{\rm th}x\right) \times f_{\gamma_{\rm SE}}(x)dx \\ &= \int_0^{+\infty} F_{\gamma_{\rm SD}}\left(\tfrac{\tilde{\gamma}_{\rm th}}{\Psi} + \gamma_{\rm th}x\right) \times f_{\gamma_{\rm SE}}(x)dx. \end{aligned} \tag{31}$$

By using (10) and (11), (31) can be found by

$$\begin{aligned} P_{\rm out}^{L} &= 1 - \tfrac{\lambda_{\rm SE}}{\Psi}\exp\left(-\tfrac{\tilde{\gamma}_{\rm th}\lambda_{\rm SD}}{\Psi}\right) \\ &\times \int_0^{+\infty} \exp\left(-x\left(\tfrac{\gamma_{\rm th}\lambda_{\rm SD}}{\Psi} + \tfrac{\lambda_{\rm SE}}{\Psi}\right)\right)dx \\ &= 1 - \tfrac{\lambda_{\rm SE}}{\gamma_{\rm th}\lambda_{\rm SD} + \lambda_{\rm SE}}\exp\left(-\tfrac{\tilde{\gamma}_{\rm th}\lambda_{\rm SD}}{\Psi}\right). \end{aligned} \tag{32}$$

3.2. Average SBLER Analysis

From (29), to derive the closed-form analytical expression, we condition the aforementioned probability on the eavesdropper's effective channel realization, denoted as y. This allows the conditional error probability to be expressed via the CDF as $F_{\gamma_{\rm D}}(\omega(1+y)-1)$. Next, averaging this term over the PDF of y yields the integral $\int_0^{+\infty} F_{\gamma_{\rm D}}(\omega(1+y)-1) f_{\gamma_{\rm E}}(y)dy$. Then, applying integration by parts rigorously transforms this integral into $1 - \int_0^{+\infty} F_{\gamma_{\rm E}}(y) f_{\gamma_{\rm D}}(\omega(1+y)-1)\omega\, dy$.

Finally, taking the expectation over the backscatter's channel variable x leads to the analytical average SBLER as:

$$\begin{aligned} &\varepsilon_{\rm D} = 1 - \omega\int_{\tau_{\rm EH}}^{+\infty} f_{\gamma_{\rm SB}}(x) \\ &\times \left\{\int_0^{+\infty} F_{\gamma_{\rm E}}(y) f_{\gamma_{\rm D}}(\omega(1+y)-1)dydx.\right\} \end{aligned} \tag{33}$$

Next, based on (19), the CDF of $\gamma_{\rm E}$ and $\gamma_{\rm D}$ can be expressed, respectively.

$$\begin{aligned} F_{\gamma_{\rm E}}(y) &= \Pr(\gamma_{\rm E} < y) = 1 - \Pr(\gamma_{\rm E} \ge y) \\ &= 1 - \Pr\left(\gamma_{\rm SE} \ge \beta xy\gamma_{\rm BE}, \gamma_{\rm BE} \ge \tfrac{y}{\beta\Psi x}\right) \\ &= 1 - \int_{\frac{y}{\beta\Psi x}}^{+\infty} \Pr(\gamma_{\rm SE} \ge \beta xyv) f_{\gamma_{\rm BE}}(v)dv \\ &= 1 - \int_{\frac{y}{\beta\Psi x}}^{+\infty} \exp(-\lambda_{\rm SE}\beta xyv)\lambda_{\rm BE}\exp(-\lambda_{\rm BE}v)dv \\ &= 1 - \tfrac{\lambda_{\rm BE}}{\lambda_{\rm BE} + \lambda_{\rm SE}\beta xy}\exp\left(-\tfrac{\lambda_{\rm BE}y}{\beta\Psi x} - \tfrac{\lambda_{\rm SE}y^2}{\Psi}\right). \end{aligned} \tag{34}$$

And

$$F_{\gamma_{\rm D}}(z) = 1 - \frac{\lambda_{\rm BD}}{\lambda_{\rm BD} + \lambda_{\rm SD}\beta xz}\exp\left(-\frac{\lambda_{\rm BD}z}{\beta\Psi x} - \frac{\lambda_{\rm SD}z^2}{\Psi}\right). \tag{35}$$

Consequently, the PDF of $\gamma_{\rm D}$ can be created by

$$f_{\gamma_{\rm D}}(z) = \frac{\partial F_{\gamma_{\rm D}}(z)}{\partial z} = \frac{\lambda_{\rm BD}\exp\left(-\frac{\lambda_{\rm BD}z}{\beta\Psi x}-\frac{\lambda_{\rm SD}z^2}{\Psi}\right)}{\lambda_{\rm BD}+\lambda_{\rm SD}\beta xz} \times\left[\frac{\lambda_{\rm SD}\beta x}{\lambda_{\rm BD}+\lambda_{\rm SD}\beta xz}+\frac{\lambda_{\rm BD}}{\beta\Psi x}+\frac{2\lambda_{\rm SD}z}{\Psi}\right]. \quad (36)$$

By substituting (10), (34), and (36)(by setting $z = \omega(1+y)-1$) into (33), the average SBLER is rewritten as:

$$\mathcal{E}_{\rm D} = 1-\omega\int_{\tau_{\rm EH}}^{+\infty}\lambda_{\rm SB}e^{-\lambda_{\rm SB}x}\int_0^{+\infty}\Xi(y)dydx. \quad (37)$$

where

$$\Xi(y) = \left\{1-\frac{\lambda_{\rm BE}\exp\left(-\frac{\lambda_{\rm BE}y}{\beta\Psi x}-\frac{\lambda_{\rm SE}y^2}{\Psi}\right)}{\lambda_{\rm BE}+\lambda_{\rm SE}\beta xy}\right\} \times\frac{\lambda_{\rm BD}\exp\left(-\frac{\lambda_{\rm BD}(\omega y+\omega-1)}{\beta\Psi x}-\frac{\lambda_{\rm SD}(\omega y+\omega-1)^2}{\Psi}\right)}{\lambda_{\rm BD}+\lambda_{\rm SD}\beta x(\omega y+\omega-1)} \times\left\{\frac{\lambda_{\rm SD}\beta x}{\lambda_{\rm BD}+\lambda_{\rm SD}\beta x(\omega y+\omega-1)}+\frac{\lambda_{\rm BD}}{\beta\Psi x}+\frac{2\lambda_{\rm SD}(\omega y+\omega-1)}{\Psi}\right\}. \quad (38)$$

Observing (37), it is clear that finding a closed-form expression for the average SBLER is extremely challenging because the integral involved is highly complex. Therefore, the approximation form using Gaussian-Chebyshev quadrature [30,31] can be applied as follows

$$\begin{aligned}&\int_0^{+\infty}\Xi(y)dy \approx \frac{\pi^2}{4N}\sum_{n=1}^{N}\sqrt{1-\theta_n^2}\Xi\left(\tan\left((\theta_n+1)\tfrac{\pi}{4}\right)\right)\\ &\times\sec^2\left((\theta_n+1)\tfrac{\pi}{4}\right)\\ &=\frac{\pi^2}{4N}\sum_{n=1}^{N}\sqrt{1-\theta_n^2}\Xi(y_n)\sec^2\left((\theta_n+1)\tfrac{\pi}{4}\right),\end{aligned} \quad (39)$$

where $\theta_n = \cos\left(\frac{2n-1}{2N}\pi\right)$, $n = 1,2,...,N$, and $y_n = \tan\left((\phi_n+1)\frac{\pi}{4}\right)$.

By replacing (39) into (37), we have:

$$\mathcal{E}_{\rm D} = 1-\omega\int_{\tau_{\rm EH}}^{+\infty}\Lambda(x)dx, \quad (40)$$

where

$$\Lambda(x) = \frac{\pi^2}{4N}\sum_{n=1}^{N}\sqrt{1-\theta_n^2}\sec^2\left((\theta_n+1)\tfrac{\pi}{4}\right) \times\left[1-\frac{\lambda_{\rm BE}\exp\left(-\frac{\lambda_{\rm BE}y_n}{\beta\Psi x}-\frac{\lambda_{\rm SE}y_n^2}{\Psi}\right)}{\lambda_{\rm BE}+\lambda_{\rm SE}\beta xy_n}\right] \times\frac{\lambda_{\rm BD}\exp\left(-\frac{\lambda_{\rm BD}(\omega y_n+\omega-1)}{\beta\Psi x}-\frac{\lambda_{\rm SD}(\omega y_n+\omega-1)^2}{\Psi}\right)}{\lambda_{\rm BD}+\lambda_{\rm SD}\beta x(\omega y_n+\omega-1)} \times\left[\frac{\lambda_{\rm SD}\beta x}{\lambda_{\rm BD}+\lambda_{\rm SD}\beta x(\omega y_n+\omega-1)}+\frac{\lambda_{\rm BD}}{\beta\Psi x}+\frac{2\lambda_{\rm SD}(\omega y_n+\omega-1)}{\Psi}\right]. \quad (41)$$

Finally, continuing to apply the Gaussian-Chebyshev approximation again, the average SBLER can be approximated as

$$\mathcal{E}_{\rm D} \approx 1-\frac{\omega\pi^4}{16MN}\sum_{m=1}^{M}\sum_{n=1}^{N}\lambda_{\rm SB}e^{-\lambda_{\rm SB}x_m}\sqrt{1-\vartheta_m^2}\sqrt{1-\theta_n^2} \times\sec^2\left(\tfrac{\pi}{4}(\vartheta_m+1)\right)\sec^2\left(\tfrac{\pi}{4}(\theta_n+1)\right) \times\left[1-\frac{\lambda_{\rm BE}\exp\left(-\frac{\lambda_{\rm BE}y_n}{\beta\Psi x_m}-\frac{\lambda_{\rm SE}y_n^2}{\Psi}\right)}{\lambda_{\rm BE}+\lambda_{\rm SE}\beta x_m y_n}\right] \times\frac{\lambda_{\rm BD}\exp\left(-\frac{\lambda_{\rm BD}(\omega y_n+\omega-1)}{\beta\Psi x_m}-\frac{\lambda_{\rm SD}(\omega y_n+\omega-1)^2}{\Psi}\right)}{\lambda_{\rm BD}+\lambda_{\rm SD}\beta x_m(\omega y_n+\omega-1)} \times\left[\frac{\lambda_{\rm SD}\beta x_m}{\lambda_{\rm BD}+\lambda_{\rm SD}\beta x_m(\omega y_n+\omega-1)}+\frac{\lambda_{\rm BD}}{\beta\Psi x_m}+\frac{2\lambda_{\rm SD}(\omega y_n+\omega-1)}{\Psi}\right], \quad (42)$$

where $\vartheta_m = \cos\left(\frac{2m-1}{2M}\pi\right)$, $m = 1,2,...,M$, and $x_m = \tau_{\rm EH}+\tan\left(\frac{\pi}{4}(\vartheta_m+1)\right)$.

4. Numerical Results and Discussions

In this section, we straightforwardly validate the analytical expressions using Monte-Carlo simulations [32, 33]. These simulations serve purely as a numerical benchmark to check the accuracy of the derived closed-form and quadrature-based expressions. Each simulated point comes from averaging a large number of independent block-Rayleigh fading realizations, ensuring the simulation results stabilize and align closely with the analytical curves. The main simulation parameters are listed in the Table 1.

Tab. 1: The main simulation parameters

Parameter (Notation)	**Value**
Average rechargeable power (ϕ)	-5 dB
Monte Carlo realizations	10^6
Average transmit power of source (Ψ)	$-10 \to 20$ dB
Target rate of long-packet transmission ($R_{\rm S}^L$)	0.5, 1, 1.5 bps/Hz
Reflection coefficient (β)	0.5
Rate parameter of $\gamma_{\rm SB}$ ($\lambda_{\rm SB}$)	0.5
Rate parameter of $\gamma_{\rm BD}$ ($\lambda_{\rm BD}$)	0.5
Rate parameter of $\gamma_{\rm BE}$ ($\lambda_{\rm BE}$)	1.5
Information bits (B)	100
Length of channel uses (L)	100, 200, 300
Number of quadrature points ($M = N$)	100
Information-leakage constraint (φ)	10^{-3}

Fig. 2 illustrates the secrecy outage probability (SOP) as a function of the transmit SNR, Ψ, for different target secrecy rates, $R_{\rm L}^{\rm S}$. As expected, the SOP decreases steadily as Ψ increases. This is because a higher

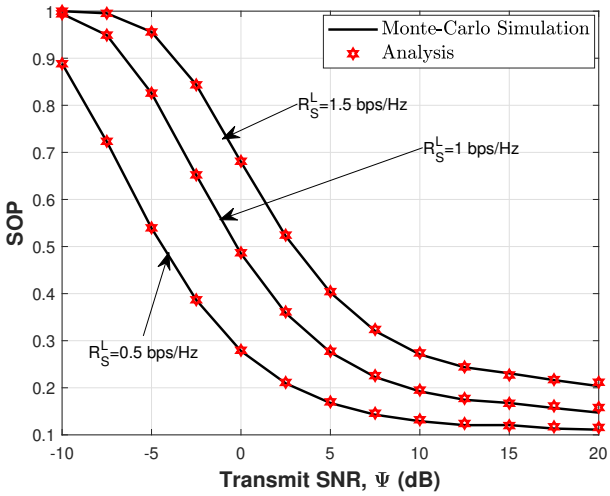


Fig. 2: SOP versus Ψ with $\phi = -5dB$, $\beta = 0.5$, $\lambda_{\rm SD}$=0.5, $\lambda_{\rm SE}$=5.5, and different $R_{\rm S}^L$=[0.5,1,1.5] bps/Hz.

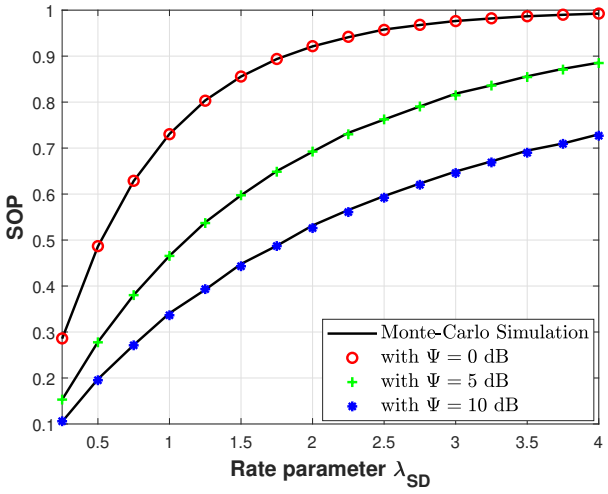


Fig. 3: SOP versus $\lambda_{\rm SD}$ with $\phi = -5dB$, $\beta = 0.5$, $\lambda_{\rm SE}$=5.5, $R_{\rm S}^L$=1 bps/Hz, and different $\Psi = 0, 5, 10dB$.

transmit SNR strengthens the legitimate $S \to D$ link and increases the likelihood that the instantaneous secrecy capacity satisfies the prescribed rate requirement. The improvement is particularly noticeable in the low- and moderate-SNR regions, where receiver noise remains a major cause of secrecy outage. Furthermore, a lower target secrecy rate consistently provides better outage performance. In particular, $R_{\rm L}^{\rm S} = 0.5$ bps/Hz yields the lowest SOP, whereas $R_{\rm L}^{\rm S} = 1.5$ bps/Hz results in the highest SOP over the entire SNR range. This behaviour can be explained by the secrecy threshold $\gamma_{\rm th} = 2^{R_{\rm L}^{\rm S}}$. Increasing $R_{\rm L}^{\rm S}$ raises $\gamma_{\rm th}$, thereby making the secrecy condition in (15) more difficult to satisfy. It can also be observed that the SOP curves approach non-zero floors in the high-SNR region. This result is consistent with the analytical expression in (32). Specifically, as $\Psi \to \infty$, the exponential term approaches unity, and the asymptotic SOP is given by $P_{\rm out}^{\rm L,\infty} = 1 - \frac{\lambda_{\rm SE}}{\gamma_{\rm th}\lambda_{\rm SD}+\lambda_{\rm SE}}$.

Therefore, increasing the source transmit power alone cannot completely eliminate secrecy outage because it simultaneously strengthens the signals received through both the legitimate and eavesdropping links. For the parameters considered in Fig. 2, the asymptotic SOP floors are approximately 0.114, 0.154, and 0.204 for $R_{\rm L}^{\rm S} = 0.5$, 1, and 1.5 bps/Hz, respectively. These results demonstrate that the target secrecy rate should be selected carefully. Although a higher target rate supports a greater confidential transmission rate, it also causes a clear reduction in secrecy reliability. Consequently, further performance improvements in the high-SNR regime require techniques that enhance the legitimate channel relative to the wiretap channel, rather than merely increasing the source transmit power.

Fig. 3 examines the effect of the legitimate-link rate parameter $\lambda_{\rm SD}$ on the SOP for different transmit SNR values. It should be recalled that, under the adopted Rayleigh fading model, the average channel power gain is $\Omega_{\rm SD} = 1/\lambda_{\rm SD}$. Hence, a larger λ_{SD} represents a weaker average $S \to D$ channel. As a result, the SOP increases monotonically with $\lambda_{\rm SD}$ because the legitimate destination becomes less likely to support the prescribed secrecy rate. This trend is also evident from (32), where an increase in $\lambda_{\rm SD}$ reduces both the exponential term and the relative advantage of the legitimate link over the wiretap link. For any given value of $\lambda_{\rm SD}$, increasing Ψ reduces the SOP. A higher transmit SNR mitigates the influence of receiver noise and therefore improves the instantaneous secrecy capacity. Nevertheless, the improvement obtained by increasing Ψ becomes less pronounced when $\lambda_{\rm SD}$ is large. In this region, the poor average quality of the legitimate channel becomes the principal limitation, and an increase in transmit power cannot fully restore the secrecy advantage of the destination. Furthermore, even when Ψ is high, the SOP does not approach zero because the eavesdropper also benefits from the increased source power. These observations indicate that improving the legitimate-channel quality, for example through favourable node placement or link selection, can be more effective than relying solely on higher transmit power.

Fig. 4 illustrates the SOP against the eavesdropping-link rate parameter $\lambda_{\rm SE}$. In contrast to the behaviour observed in Fig. 4, the SOP decreases as $\lambda_{\rm SE}$ increases. Since $\Omega_{\rm SE} = 1/\lambda_{\rm SE}$, a larger $\lambda_{\rm SE}$ corresponds to a smaller average channel power gain for the $S \to E$ link. The eavesdropper consequently receives a weaker signal, which increases the difference between the capacities of the legitimate and wiretap channels and re-

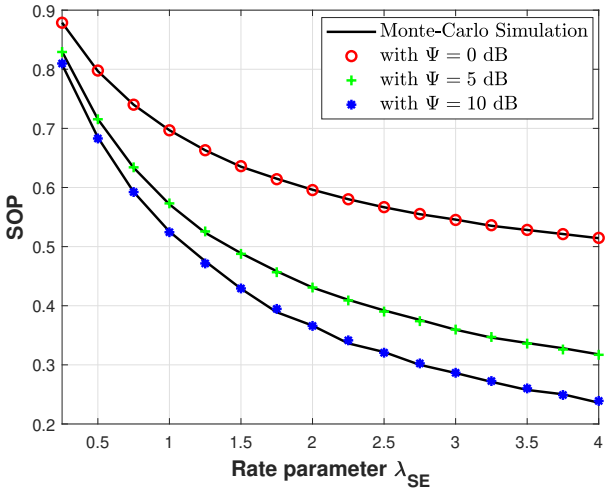


Fig. 4: SOP versus λ_{SE} with $\phi = -5dB$, $\beta = 0.5$, λ_{SD}=0.5, R_{S}^{L}=1 bps/Hz, and different $\Psi = 0, 5, 10dB$.

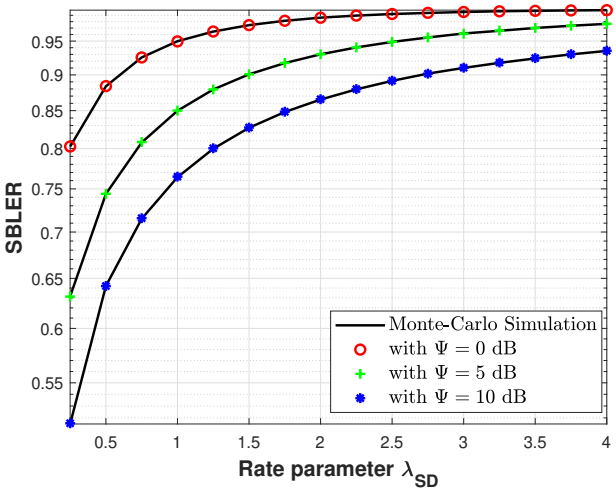


Fig. 6: SBLER versus λ_{SD} with $\phi = -5dB$, $\beta = 0.5$, λ_{SB}=λ_{BD}=0.5, λ_{SE}=5.5, λ_{BE}=1.5, $B = 100$, $L = 200$, and different $\Psi = 0, 5, 10dB$.

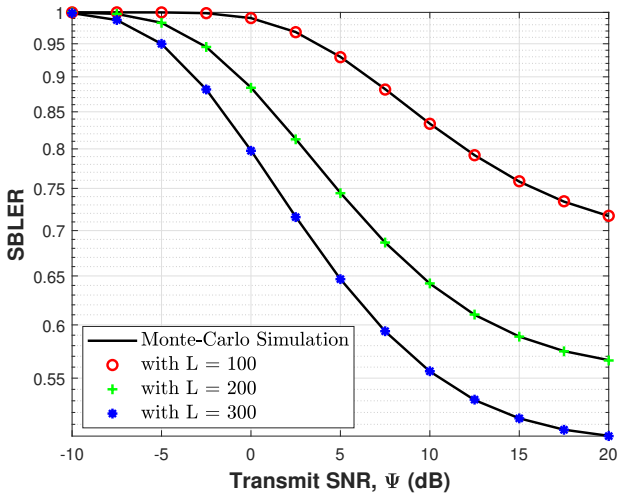


Fig. 5: SBLER versus Ψ with $\phi = -5dB$, $\beta = 0.5$, λ_{SD}=λ_{SB}=λ_{BD}=0.5, λ_{SE}=5.5, λ_{BE}=1.5, $B = 100$ and different L=[100, 200, 300].

duces the probability of secrecy outage. This result is consistent with (32), in which increasing λ_{SE} increases the fraction $\frac{\lambda_{\mathrm{SE}}}{\gamma_{\mathrm{th}}\lambda_{\mathrm{SD}}+\lambda_{\mathrm{SE}}}$, thereby lowering $P_{\mathrm{out}}^{\mathrm{L}}$. The results also show that a larger Ψ provides lower SOP values throughout the range of λ_{SE}. The improvement is especially relevant when the eavesdropping channel is relatively strong, namely when λ_{SE} is small, because the system must overcome both receiver noise and an unfavourable wiretap link. As λ_{SE} increases, the eavesdropper becomes progressively disadvantaged, and the SOP is governed mainly by the legitimate-link condition. Therefore, Fig. 4 confirms that secrecy performance depends not only on the absolute strength of the legitimate link, but also on its relative advantage over the eavesdropping link.

Fig. 5 presents the average SBLER as a function of the transmit SNR for different blocklengths L, while the information payload is fixed at $B = 100$ bits. The SBLER decreases as Ψ increases because a higher source power produces two beneficial effects. First, it reduces the energy-activation threshold $\tau_{\mathrm{EH}} = \frac{\phi}{\Psi(1-\beta)}$, which lowers the probability that the backscatter device remains inactive. Second, it strengthens the backscattered short-packet signal, thereby improving the effective SINR at the legitimate destination. These effects reduce both the energy-outage and information-outage components included in (29). For a fixed transmit SNR, increasing L improves secure transmission reliability and yields a lower SBLER. Since the payload B is held constant, a larger blocklength reduces the coding rate $R_{\mathrm{S}} = B/L$. It also decreases the finite-blocklength dispersion penalties in (21), which scale with $1/\sqrt{L}$. The destination therefore has a greater coding margin with which to satisfy the reliability and information-leakage requirements. The relative improvement due to a longer blocklength becomes more evident once the energy-outage probability has been sufficiently reduced. In the low-SNR region, by comparison, all blocklength configurations exhibit a high SBLER because insufficient harvested energy and poor received signal quality remain the dominant causes of failure. Fig. 5 thus demonstrates the fundamental reliability–latency trade-off: increasing L enhances secure reliability, but it also requires a longer transmission interval.

Fig. 6 shows how the SBLER varies with the legitimate-link rate parameter λ_{SD} for several transmit SNR values. The SBLER rises with λ_{SD} because a larger rate parameter corresponds to a weaker average $S \to D$ channel. According to (19), successful recovery

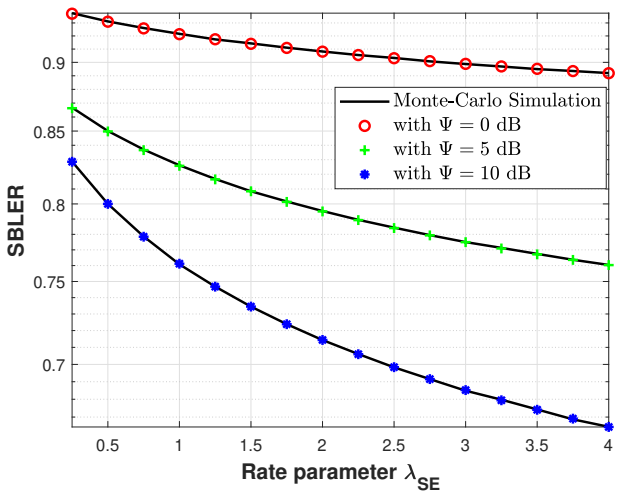


Fig. 7: SBLER versus λ_{SE} with $\phi = -5dB$, $\beta = 0.5$, λ_{SD}=λ_{SB}=λ_{BD}=0.5, λ_{BE}=1.5, $B = 100$, $L = 200$, and different $\Psi = 0, 5, 10dB$.

of the short packet requires the destination to complete both stages of successive decoding. In particular, it must first decode the long-packet component and then recover the backscattered short packet after cancellation. Since the effective SINR γ_{D} is determined by the weaker of these two decoding stages, deterioration of the direct legitimate link increases the probability that the overall decoding condition will not be satisfied. Increasing Ψ reduces the SBLER for a given λ_{SD} by raising the received signal strength and lowering the probability of an energy outage at the backscatter device. However, the benefit of a higher transmit SNR diminishes as λ_{SD} becomes large. In this case, the quality of the direct legitimate link limits the first decoding stage, and additional source power cannot fully compensate for severe fading. The figure therefore highlights the importance of maintaining a reliable $S \to D$ link, even though the short packet itself is conveyed through the cascaded $S \to BD \to D$ path.

Finally, Fig. 7 depicts the SBLER as a function of the wiretap-link rate parameter λ_{SE}. The SBLER decreases as λ_{SE} increases because a larger λ_{SE} weakens the average $S \to E$ channel. This reduces the eavesdropper's effective SINR γ_{E} and increases the probability that $\gamma_{\mathrm{D}} > \gamma_{\mathrm{E}}$, which is required to support a positive secrecy rate. A weaker wiretap link also increases the instantaneous secrecy capacity in (21), making it easier for the short-packet transmission to satisfy both the target coding rate and the prescribed information-leakage constraint. For each value of λ_{SE}, increasing Ψ generally improves the SBLER performance because it facilitates the activation of the backscatter device and strengthens the link to the legitimate destination. Nevertheless, the improvement due to transmit power is limited by the fact that the eavesdropper also receives a stronger source signal. Consequently, the relative channel advantage between the destination and the eavesdropper remains a decisive factor, particularly at moderate and high SNR values. Taken together, Figs. 6 and 7 show that secure short-packet reliability is highly sensitive to the balance between the legitimate and eavesdropping links. Improving the destination's channel or weakening the wiretap channel can therefore provide a more substantial benefit than increasing the source power alone.

5. Conclusion

This paper has analysed the secrecy performance of an energy-harvesting backscatter-assisted system supporting hybrid long- and short-packet transmissions over independent Rayleigh fading channels. A closed-form expression was derived for the secrecy outage probability of the long-packet transmission. For secure short-packet transmission, the average secure block error rate was formulated by jointly considering the energy-activation condition, successive decoding, finite-blocklength coding, and information leakage. An efficient Gaussian–Chebyshev quadrature approximation was then developed to evaluate the resulting intractable integrals. The numerical results confirmed the accuracy of the analytical framework. Increasing the transmit SNR generally improved secrecy and short-packet reliability by reducing the effect of noise and the probability of energy outage. Nevertheless, transmit-power enhancement alone could not eliminate secrecy failures because it also strengthened the signal received by the eavesdropper. The results further showed that the relative qualities of the legitimate and wiretap links played a decisive role in both SOP and SBLER performance. Moreover, increasing the blocklength reduced the SBLER by lowering the coding rate and finite-blocklength penalty, although this improvement was accompanied by increased transmission latency.

Future work may consider imperfect channel knowledge, imperfect interference cancellation, non-linear energy harvesting, and joint optimisation of the reflection coefficient and packet length.

Author Contributions

Quang-Sang Nguyen and Minh Tran contributed to the conceptualisation, system modelling, mathematical analysis, and numerical simulations. Do Hoang Minh Han and Bui Vu Minh contributed to the preparation of the original manuscript. Cong-Hung Tran and Minh Tran contributed to the validation of the analytical results and the critical revision of the manuscript. Rupak

Kharel contributed to the supervision of the study and the review and editing of the manuscript. All authors discussed the results, reviewed the manuscript, and approved the final version.

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